

Compact laser displacement sensor
CD22 Series

Quick Start



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What does this quick start cover?



This quick start explains the basic settings required to measure and detect objects.



Setting the analog output

Analog output type only

Use this procedure to set the points at which to generate the minimum and maximum analog output and the directions for increasing and decreasing this output.



Setting the control output thresholds (the switching references)

Use this procedure to set the thresholds used as the references for switching control output ON and OFF.

- 1 Selecting the teach mode (threshold setting type)
- 2 Executing the teach (setting the thresholds)
- 3 Fine adjustment the thresholds



For information not included in this guide (such as installation, wiring, specifications, and a list of the functions), see the instruction manual included with the product or the OPTEX FA website.



Setting the analog output

Analog output type only

Use this procedure to set the output range of the analog output and the directions for increasing and decreasing this output.

Displayed distance

On the CD22 series, the center point of the measurement range is displayed as "0". Distances further away from the sensor than the center point are added to the value of the center point and are displayed as positive numbers. Distances closer to the sensor than the center point are subtracted from the value of the center point and are displayed as negative numbers.

	Minimum distance	Center point of measurement range	Maximum distance
	Subtracted	"0"	Added
[Displayed values]			
CD22-15x	-5.000 mm (10 mm)	0.000 mm (15 mm)	5.000 mm (20 mm)
CD22-35x	-15.00 mm (20 mm)	0.00 mm (35 mm)	15.00 mm (50 mm)
CD22-100x	-50.00 mm (50 mm)	0.00 mm (100 mm)	50.00 mm (150 mm)

() : Measurement distance

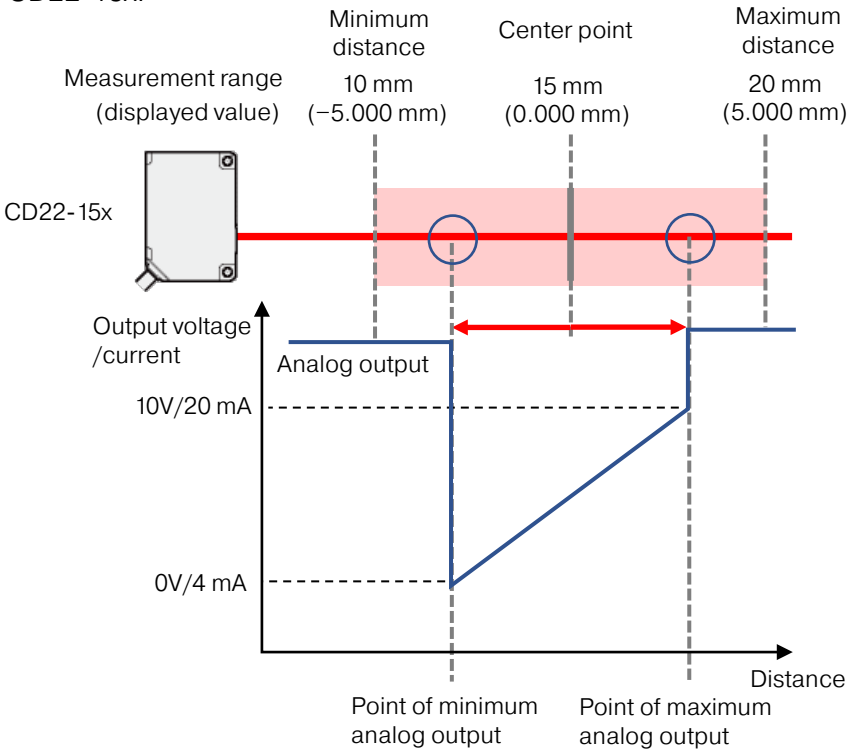


"9999" is displayed when the object is outside the measurement range or cannot be measured.

Setting the analog output

Use this procedure to set the sensing distances at which to generate the minimum (0V/4mA) and maximum (10V/20mA) analog output.

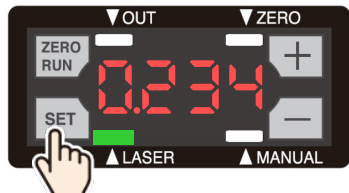
As an example, this section explains how to set the analog output on the CD22-15x.



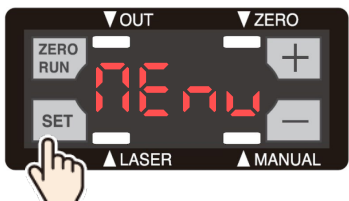
Canceling setup

Press [ZERO RUN]. The sensor returns to [RUN] mode.

- (1) In [RUN] mode (with the measured value displayed), press [SET].

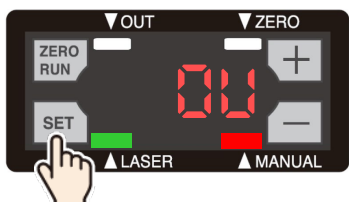


- (2) [MENU] appears. Press [SET].



- (3) The minimum analog output appears. Press [SET].

- Voltage output type: [0V]
- Current output type: [4mA]



- (4) The present minimum blinks.

Use the [+] or [-] key to enter the distance at which to generate the minimum analog output.



<Reference> Default minimum values

Model name	Minimum (0V/4mA)
CD22-15x	-5.000 mm
CD22-35x	-15.000 mm
CD22-100x	-50.00 mm

The negative sign is displayed here.

As an example, the value is set to [-2.000] here. At a distance of [-2.000], the minimum analog output ([0V] or [4mA]) is generated.

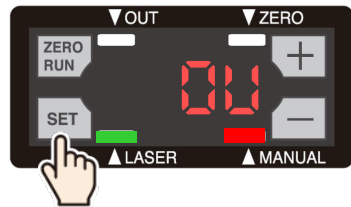
(5) Enter the distance and press [SET].
The minimum is confirmed, and the sensor returns to the previous screen.

- Voltage output type: [0V]
- Current output type: [4mA]

0V

4mA

e.g.) Analog voltage output type



(6) Set the maximum.

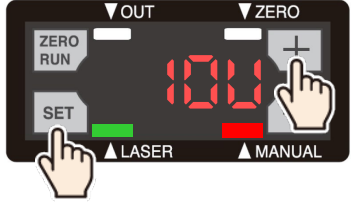
Press the [+] key to display the maximum analog output. Press [SET].

- Voltage output type: [10V]
- Current output type: [20mA]

10V

20mA

e.g.) Analog voltage output type



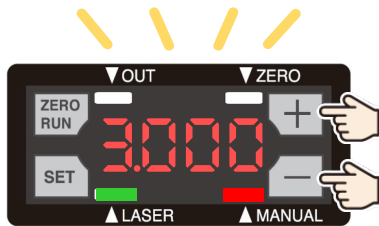
(7) The present maximum blinks.

Use the [+] or [-] key to enter the distance at which to generate the maximum analog output.

<Reference> Default maximum values

Model name	Maximum (10V/20mA)
CD22-15x	5.000 mm
CD22-35x	15.000 mm
CD22-100x	50.00 mm

As an example, the value is set to [3.000] here.
At a distance of [3.000], the maximum analog output ([10V] or [20mA]) is generated.



(8) Enter the distance and press [SET].

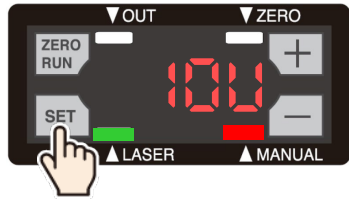
The maximum is confirmed, and the sensor returns to the previous screen.

- Voltage output type: [10V]
- Current output type: [20mA]

10V

20mA

e.g.) Analog voltage output type



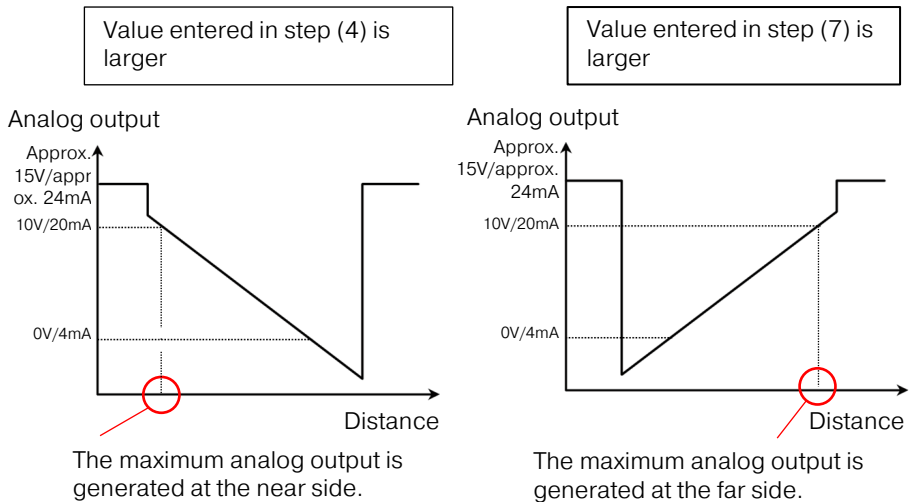
■ This completes the setup.

Press [ZERO RUN] to return the sensor to [RUN] mode.



Directions for increasing and decreasing analog output

As shown below, the directions in which analog output increases and decreases vary depending on which of the values entered in steps (4) and (7) is larger.



Setting the control output thresholds (the switching references)

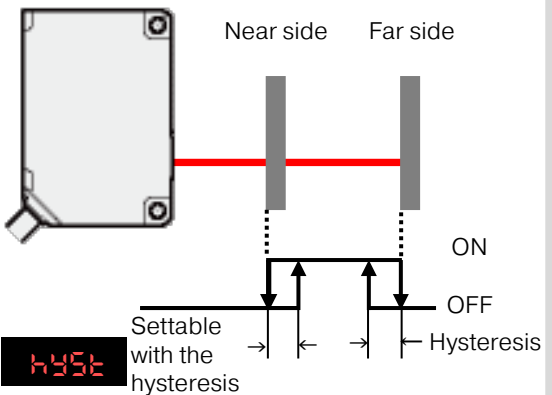
Use teach mode to set the thresholds used as the references for switching control output ON and OFF.
To match different applications, the following three modes are available.

- 2-point teach
- 1-point teach
- FGS2 teach

Output range differences depending on the teach mode

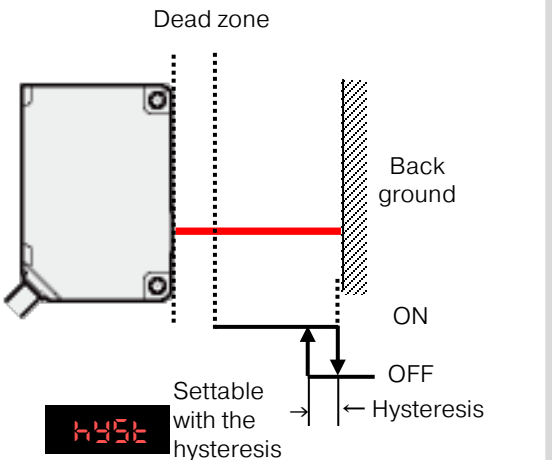
2Pt 2-point teach

Teach is executed with the object at two positions: the near and far points in relation to the sensor. Then, output turns ON when the measurement distance is between these positions.



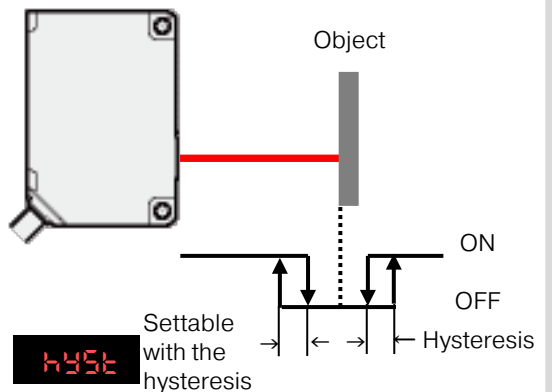
1Pt 1-point teach

Teach is executed for one position: without the object present (that is, just the background is detected). Then, the output turns ON when the measurement distance is closer than that position. (Excluding the dead zone in front of the sensor.)



FGS2 FGS2 teach

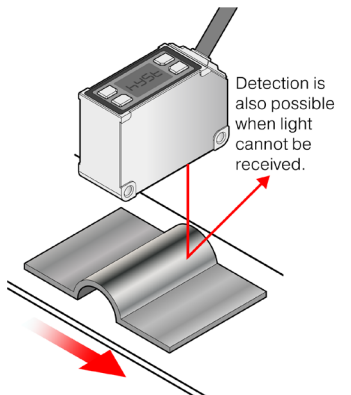
Place the object and execute FGS2 teach to set the detection range around the object. (Two thresholds are set, one before and one after the object, and the output turns OFF when the measurement distance is within this range.)



FGS2 teach features

The sensor receives light from the background (such as a white conveyor belt) at all times and activates when this light is blocked by the object, so detection is also possible when the sensor can no longer receive light. This is the optimal mode for conventionally hard to detect objects such as thin black objects, glossy metal, and objects with rough surfaces.

* FGS2 teach requires a stable background such as a conveyor belt.

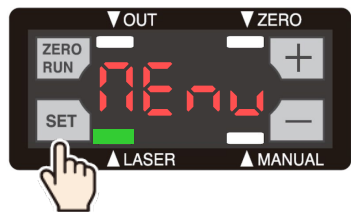


1 Selecting the teach mode

Select the teach mode before executing teach.

 **Canceling setup**
Press [ZERO RUN]. The sensor returns to [RUN] mode.

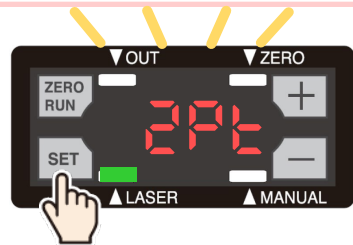
- (1) In [RUN] mode (with the measured value displayed), press [SET].
[MENU] appears.



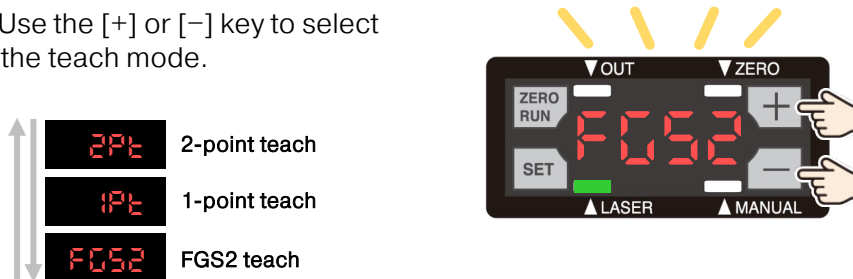
- (2) Press the [+] or [-] key multiple times until [MODE] appears.



- (3) Press [SET].
The present teach mode blinks.



- (4) Use the [+] or [-] key to select the teach mode.



Press the [+] or [-] key to scroll through the teach modes.

- This completes the selection of the teach mode.
Press [ZERO RUN] to return the sensor to [RUN] mode.



Next, set the threshold(s) for the teach mode displayed with this procedure.
To do so, refer to “2. Executing the teach mode”.

2 Executing the teach mode

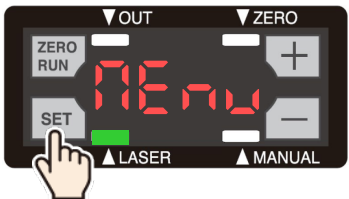
Prepare the environment and the object to use for the detection conditions, and then execute the teach mode.
This section uses the display of the CD22-15x type.

2Pt Executing 2-point teach

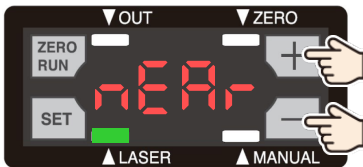
 **Before executing the teach mode**
Select 2-point teach **2Pt** in “1. Selecting the teach mode”.

 **Canceling setup**
Press [ZERO RUN]. The sensor returns to [RUN] mode.

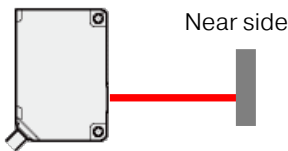
- (1) In [RUN] mode (with the measured value displayed), press [SET].
[MENU] appears.



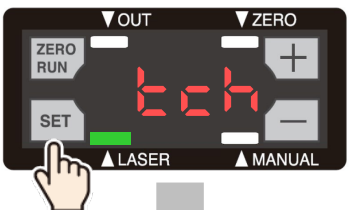
- (2) Press the [+] or [-] key multiple times until [NEAR] appears.



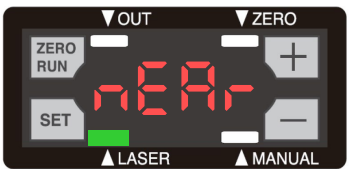
- (3) Position the object near to the sensor.



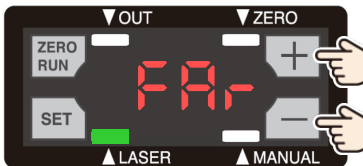
- (4) Press [SET].
[tch] appears, and the near-side threshold is set.



[NEAR] appears on the screen again.



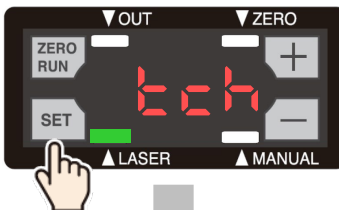
- (5) Execute the teach for the far side.
Press the [+] or [-] key multiple times until [FAR] appears.



- (6) Position the object far from the sensor.



- (7) Press [SET].
[tch] appears, and the far-side threshold is set.



[FAR] appears on the screen again.



- This completes the execution of 2-point teach.
Press [ZERO RUN] to return the sensor to [RUN] mode.



1Pt Executing 1-point teach



Before executing the teach mode

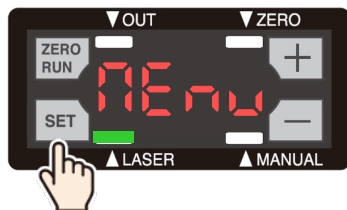
Select 1-point teach **1Pt** in “1. Selecting the teach mode”.



Canceling setup

Press [ZERO RUN]. The sensor returns to [RUN] mode.

- (1) In [RUN] mode (with the measured value displayed), press [SET].
[MENU] appears.



- (2) Press the [+] or [-] key multiple times until [FAR] appears.

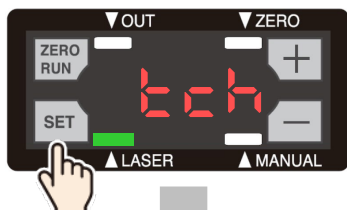


- (3) Shine light on the background (the situation where the object is not present).



- (4) Press [SET].

[tch] appears, and the far-side threshold is set.



[FAR] appears on the screen again.



- This completes the execution of 1-point teach.

Press [ZERO RUN] to return the sensor to [RUN] mode.



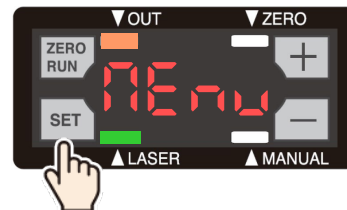
FGS2 Executing FGS2 teach



Before executing the teach mode

Select FGS teach mode **FGS2** in “1. Selecting the teach mode”.

- (1) In [RUN] mode (with the measured value displayed), press [SET].
[MENU] appears.



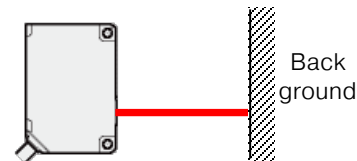
- (2) Press the [+] or [-] key multiple times until [FGS2] appears.



- (3) Prepare the situation to teach the sensor.
Perform operations (3-1) and (3-2) below if necessary for the application.

(3-1) To detect when the sensor does not receive light.

(Refer to “FGS2 teach mode features” on page 2.)

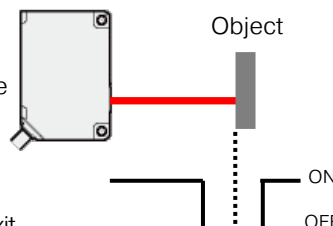


Shine light on the background (the situation where the object is not present).

(3-2) To set the detection range around the object

Note: By default, the detection range indicates the range in which the output is OFF.
To change this so the detection range indicates the range in which the output is ON, fully carry out FGS2 teach, and then exit teach mode.

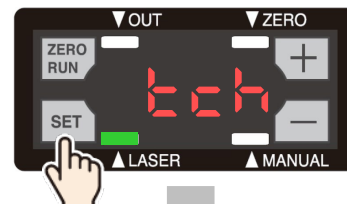
Then, refer to the instruction manual and set **Act** to **don** (set the output polarity to Dark ON) in Setup mode.



Place the object in the middle of the detection range.

- (4) Press [SET].

[tch] appears, and the detection range is set.



[FGS2] appears on the screen again.



- This completes the execution of FGS2 teach.

Press [ZERO RUN] to return the sensor to [RUN] mode.



<Reference> Default FGS2 detection ranges

Model name	Detection range corresponding to taught distance
CD22-15x	±1.000 mm
CD22-35x	±03.00 mm
CD22-100x	±10.00 mm



To change the detection range, refer to the instruction manual and set “6: FGS2 hysteresis” **tol** (tolerance) in Setup mode.



You can fine adjustment the distance to the center of the detection range set with FGS2 teach.
To do so, refer to the next section, “Fine adjustment the thresholds (entering numeric values)”.

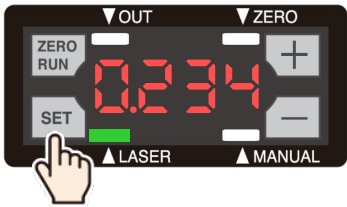
3 Fine adjustment the thresholds (entering numeric values)

You can fine-adjust the thresholds set in teach mode (the distance to the center of the detection range for FGS2 teach) by entering numeric values.

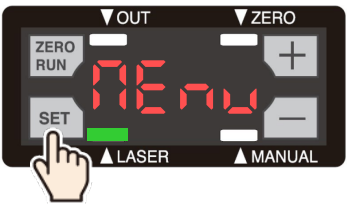
The distances set here are from the center of the measurement range. Refer to “Displayed distance” on page 1 to set these values.

2Pt Fine adjustment the 2-point teach results

(1) In [RUN] mode (with the measured value displayed), press [SET].



(2) [MENU] appears. Press [SET].



(3) The minimum analog output

Analog output type only

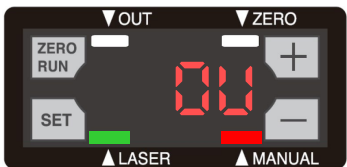
• Voltage output type: [0V]



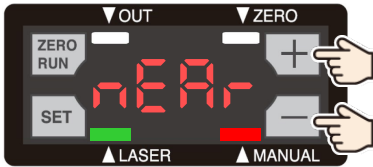
• Current output type: [4mA]



e.g.) Analog voltage output type

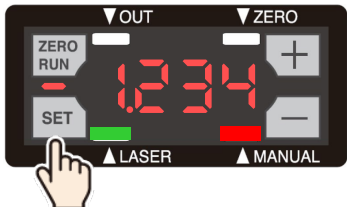


(4) Press the [+] or [-] key multiple times until [NEAR] appears.



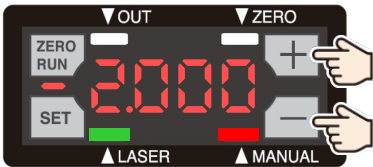
(5) Press [SET].

The near-side threshold appears.



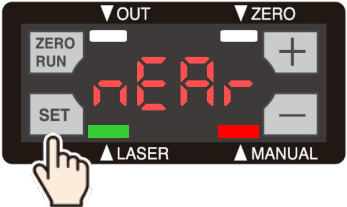
(6) Use the [+] or [-] key to set the threshold.

As an example, the value is set to [-2.000] here.

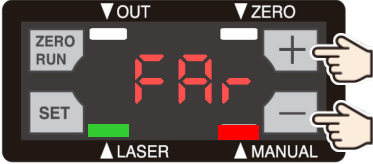


(7) Press [SET].

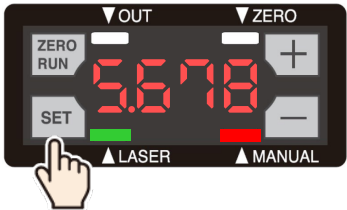
[NEAR] appears on the screen again.



(8) Press the [+] or [-] key multiple times until [FAR] appears.

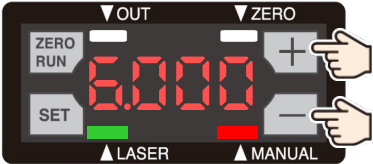


(9) Press [SET].
The far-side threshold appears.



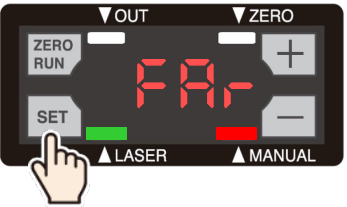
(10) Use the [+] or [-] key to set the threshold.

As an example, the value is set to [6.000] here.



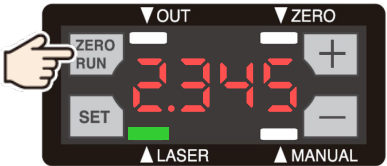
(11) Press [SET].

[FAR] appears on the screen again.



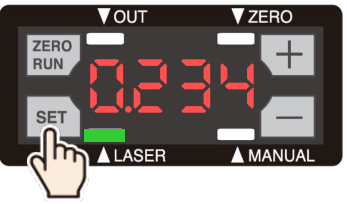
This completes the fine adjustment of the 2-point thresholds.

Press [ZERO RUN] to return the sensor to [RUN] mode.

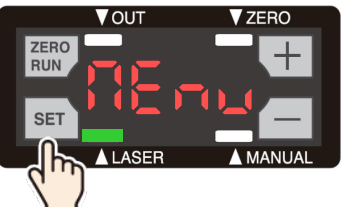


1Pt Fine adjustment the 1-point teach results

(1) In [RUN] mode (with the measured value displayed), press [SET].



(2) [MENU] appears. Press [SET].



(3) The minimum analog output

Analog output type only

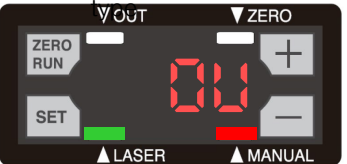
• Voltage output type: [0V]



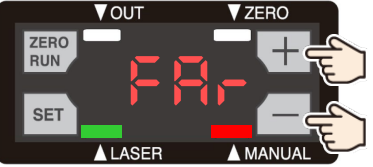
• Current output type: [4mA]



e.g.) Analog voltage output

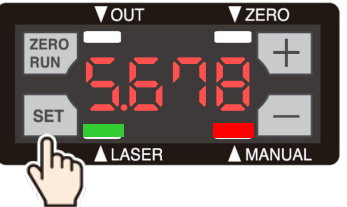


(4) Press the [+] or [-] key multiple times until [FAR] appears.

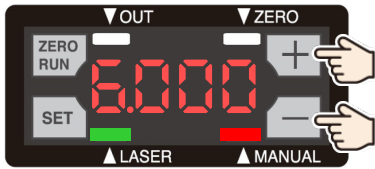


(5) Press [SET].

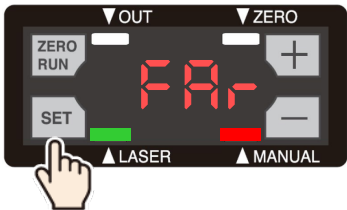
The far-side threshold appears.



(6) Use the [+] or [-] key to set the threshold.
As an example, the value is set to [6.000] here.



(7) Press [SET].
[FAR] appears on the screen again.



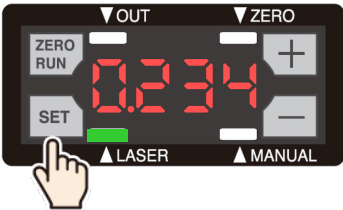
■ This completes the fine adjustment of the threshold.
Press [ZERO RUN] to return the sensor to [RUN] mode.



Note: [NEAR] (the point of minimum analog output) cannot be set with 1-point teach.

FGS2 Fine adjustment the FGS2 teach results

(1) In [RUN] mode (with the measured value displayed), press [SET].



(2) [MENU] appears. Press [SET].



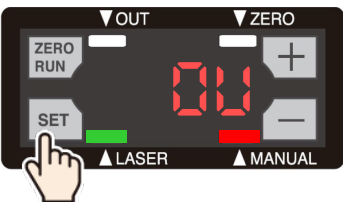
(3) The minimum analog output

Analog output type only

• Voltage output type: [0V]
• Current output type: [4mA]



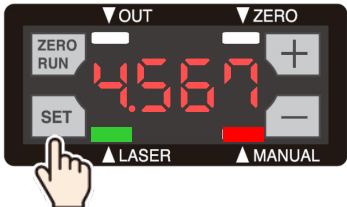
e.g.) Analog voltage output type



(4) Press the [+] or [-] key multiple times until [FGS2] appears.



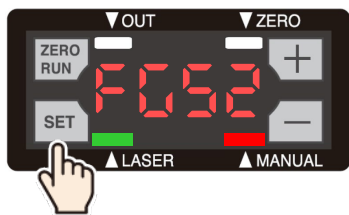
(5) Press [SET].
The distance to the center of the detection range appears.



(6) Use the [+] or [-] key to set the threshold.
As an example, the value is set to [5.000] here.



(7) Press [SET].
[FGS2] appears on the screen again.



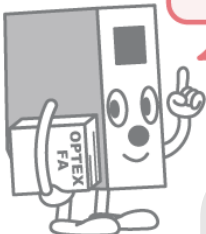
■ This completes the fine adjustment of the distance to the center of the detection range.

Press [ZERO RUN] to return the sensor to [RUN] mode.



Note: [NEAR] and [FAR] cannot be set with FGS2 teach. Even if you set them, the settings will be discarded.
To change the detection range, refer to the instruction manual and set 6: FGS2 hysteresis (tolerance) in Setup mode.

You're ready to go!



This completes the explanation provided by this quick start .

For the installation, specifications, and a list of the functions of the sensor, see the instruction manual included with the product.

